

ASSIGNMENT 4

ROLL NUMBER : 202510605

Q1. C++ Project Assignment. Simple Notepad Application

Main.cpp

```
#include "notepad.h"
```

```
int main() {
    int choice;
    string filename;
    cout << "    WELCOME TO SIMPLE NOTEPAD C++    " << endl;

    do {
        cout << "\n===== SIMPLE NOTEPAD =====" << endl;
        cout << " 1. Write File (Create / Overwrite)" << endl;
        cout << " 2. Read File (Display Contents) " << endl;
        cout << " 3. Append File (Add to File)    " << endl;
        cout << " 4. Exit                " << endl;
        cout << "Enter your choice (1-4): ";
        cin >> choice;
        cin.ignore();

        switch (choice) {
            case 1:
                cout << "\nEnter filename to write to: ";
                getline(cin, filename);
                writeFile(filename);
                break;

            case 2:
                cout << "\nEnter filename to read: ";
                getline(cin, filename);
                readFile(filename);
                break;

            case 3:
                cout << "\nEnter filename to append to: ";
                getline(cin, filename);
                appendFile(filename);
                break;

            case 4:
                cout << "\nGoodbye!" << endl;
                break;
        }
    } while (choice != 4);
}
```

```

        default:
            cout << "\n[WARNING] Invalid choice. Enter a number from 1 to 4." << endl;
            break;
    }

} while (choice != 4);

return 0;
}

```

Notepad.h

```

#ifndef NOTEPAD_H
#define NOTEPAD_H

```

```

#include <iostream>
#include <fstream>
#include <string>

```

```

using namespace std;

```

```

void writeFile(string filename);
void readFile(string filename);
void appendFile(string filename);

```

```

#endif

```

Notepad.cpp

```

#include "notepad.h"

```

```

void writeFile(string filename)

```

```

{
    ofstream file(filename);

```

```

    string text;

```

```

    cin.ignore();

```

```

    cout << "Enter text (type END to stop):" << endl;

```

```
while (true)
{
    getline(cin, text);

    if (text == "END")
    {
        break;
    }

    file << text << endl;
}

file.close();

cout << "File saved successfully." << endl;
}

void readFile(string filename)
{
    ifstream file(filename);

    string text;

    if (!file)
    {
        cout << "File not found." << endl;
        return;
    }

    cout << "\nFile Contents:" << endl;

    while (getline(file, text))
    {
        cout << text << endl;
    }
}
```

```
    }

    file.close();
}

void appendFile(string filename)
{
    ofstream file(filename, ios::app);

    string text;

    cin.ignore();

    cout << "Enter text to append (type END to stop):" << endl;

    while (true)
    {
        getline(cin, text);

        if (text == "END")
        {
            break;
        }

        file << text << endl;
    }

    file.close();

    cout << "Text appended successfully." << endl;
}
```